

If you're completing this workshop at your own pace, you can use your personal AWS account. This guide will help you set up the necessary permissions and configure AWS CLI for use with Kiro IDE.

AWS Costs

You are responsible for the cost of AWS services used while running this workshop in your AWS account. Most resources used in this workshop are eligible for the AWS Free Tier, but charges may apply depending on your account age and usage patterns.

Prerequisites

Before proceeding, ensure you have:

1. An active AWS account with console access
2. [AWS CLI](#) installed on your local machine
3. Kiro IDE installed and authenticated

Step 1: Create Workshop IAM Role

We'll use a CloudFormation template to automatically create an IAM role with the necessary permissions for this workshop.

The CloudFormation template creates

- An IAM role named `kiro-workshop-participant-role`
- `PowerUserAccess` permissions (full access to AWS services except IAM and Organizations management)
- Trust policy allowing you to assume the role from your AWS account

Download the CloudFormation Template

First, download the CloudFormation template to your local machine:

Download Template

Deploy the Stack

1. Open the [CloudFormation Console](#) in `us-east-1`.
2. Choose **Upload a template file**.
3. Click **Choose file** and select the downloaded `workshop-participant-role-template.yaml`.
4. Click **Next**.

CloudFormation > Stacks > Create stack

Step 1: Create stack

Step 2: Specify stack details

Step 3: Configure stack options

Step 4: Review and create

Create stack

Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [IaC generator](#).

Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

Choose an existing template

Upload or choose an existing template.

Build from Infrastructure Composer

Create a template using a visual builder.

Specify template

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

Amazon S3 URL

Provide an Amazon S3 URL to your template.

Upload a template file

Upload your template directly to the console.

Sync from Git

Sync a template from your Git repository.

Upload a template file

Choose file

workshop-participant-role-template.yaml

JSON or YAML, formatted file

S3 URL

<https://s3.us-east-1.amazonaws.com/cf-templates-1711nu914kc-us-east-1/2026-02-24T160805.7972yw2-workshop-participant-role-template.yaml>

[View in Infrastructure Composer](#)

Cancel

Next

5. For **Stack name**, enter: `kiro-workshop-participant-role`

CloudFormation > Stacks > Create stack

Step 1: Create stack

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Specify stack details

Provide a stack name

Stack name

kiro-workshop-participant-role

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 30/128.

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

No parameters

There are no parameters defined in your template

Cancel

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6. Click **Next** twice to reach the Review page.
7. Scroll to the bottom and check the box: **I acknowledge that AWS CloudFormation might create IAM resources**.

Capabilities

The following resource(s) require capabilities: [AWS::IAM::Role]

This template contains Identity and Access Management (IAM) resources. Check that you want to create each of these resources and that they have the minimum required permissions. In addition, they have custom names. Check that the custom names are unique within your AWS account. [Learn more](#)

☒ I acknowledge that AWS CloudFormation might create IAM resources with custom names.

Cancel

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8. Click **Submit**.
9. Wait for stack creation to complete (1-2 minutes). The status will change from `CREATE_IN_PROGRESS` to `CREATE_COMPLETE`.

Retrieve Role Information

Once the stack is created, retrieve your role details:

1. In the CloudFormation console, filter the `kiro-workshop-participant-role` stack name.
2. Click the **Outputs** tab.

CloudFormation > Stacks > kiro-workshop-participant-role

Stacks (1)

kiro-workshop-participant-role

Filter status

Active

View nested

Stacks

kiro-workshop-participant-role

2026-02-25 08:10:40 UTC+1100

CREATE_COMPLETE

kiro-workshop-participant-role

Delete

Update stack

Stack actions

Create stack

Stack info

Events

Resources

Outputs

Parameters

Template

Change sets

Git sync

Outputs (4)

Search outputs

Key	Value	Description
AssumeRoleCommand	aws sts assume-role --role-arn arn:aws:iam::[redacted]:role/kiro-workshop-participant-role --role-session-name kiro-workshop-session	AWS CLI command to assume this role
ConsoleLoginUrl	https://signin.aws.amazon.com/swi?templateRoleName=kiro-workshop-participant-role&account=[redacted]	AWS Console URL to switch to this role
RoleArn	arn:aws:iam::[redacted]:role/kiro-workshop-participant-role	IAM Role ARN (use this to assume the role)
RoleName	kiro-workshop-participant-role	IAM Role Name

3. Copy the **RoleArn** value - you'll need it in the next step. It will look like:

```
arn:aws:iam::123456789012:role/kiro-workshop-participant-role
```

What You'll See

The Outputs tab shows:

- **RoleName:** `kiro-workshop-participant-role`
- **RoleArn:** The full ARN you'll use for AWS CLI configuration
- **AssumeRoleCommand:** Ready-to-use CLI command for manual role assumption
- **ConsoleLoginUrl:** URL to switch to this role in the AWS Console

Step 2: Configure AWS CLI to Use the Role

Now configure AWS CLI to assume the workshop role. You'll need your existing AWS credentials (from your root or admin user) to assume the role.

1. Open your AWS config file in a text editor:
 - **Windows:** `C:\Users\<USERNAME>\.aws\config`
 - **Linux/Mac:** `~/.aws/config`
2. Add a new profile for the workshop role (replace `YOUR_ACCOUNT_ID` with your actual 12-digit account ID from the RoleArn):

```
[profile kiro-workshop]
role_arn = arn:aws:iam:YOUR_ACCOUNT_ID:role/kiro-workshop-participant-role
source_profile = default
region = us-east-1
output = json
```

3. Save the file.

► **Alternative: Use Environment Variables**

Step 3: Verify Configuration

1. Test your AWS CLI configuration in Kiro IDE by running:

```
aws sts get-caller-identity --profile kiro-workshop
```

2. Verify the output shows you've assumed the role. You should see:

```
{
  "UserId": "AIDACKCEVS06C2EXAMPLE:kiro-workshop-session",
  "Account": "123456789012",
  "Arn": "arn:aws:sts::123456789012:assumed-role/kiro-workshop-participant-role/kiro-workshop-session"
}
```

Configuration Complete

Your Kiro IDE is now configured to use the workshop role. You can proceed with the workshop exercises.

Note: When running AWS CLI commands in the workshop, use `--profile kiro-workshop` or set `AWS_PROFILE=kiro-workshop` as an environment variable.

Best Practices

Security Recommendations

1. **Use IAM Users, Not Root:** Never use your AWS root account credentials for daily operations
2. **Enable MFA:** Add multi-factor authentication to your IAM user for enhanced security
3. **Rotate Access Keys:** Regularly rotate your access keys (every 90 days recommended)
4. **Use Least Privilege:** Grant only the permissions needed for the workshop
5. **Monitor Usage:** Regularly check AWS Cost Explorer and CloudWatch for unexpected activity

Cost Management

1. **Set Up Billing Alerts:** Configure CloudWatch billing alarms to notify you of unexpected charges
2. **Use AWS Budgets:** Create a budget to track workshop spending
3. **Clean Up Resources:** Delete all workshop resources when finished to avoid ongoing charges
4. **Review Free Tier:** Understand which services and usage levels are covered by the Free Tier

► **Setting Up Billing Alerts**

Troubleshooting

► **AWS CLI Not Found**

► **Invalid Credentials Error**

► **Permission Denied Errors**

► **Region Issues**

Next Steps

With your personal AWS account configured:

1. Verify you can access AWS services through Kiro
2. Review the workshop prerequisites
3. Proceed to the workshop exercises

Remember

You are responsible for any AWS costs incurred. Make sure to clean up all resources at the end of the workshop to avoid ongoing charges.

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